



Registers		Flag
A	07	S 0
BC	02 00	Z 0
DE	00 00	AC 0
HL	00 00	P 0
PSW	00 00	C 0
PC	42 0C	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 8050
2 MOV B, A
3 LDA 8051
4 ADD B
5 STA 8052
6 HLT
```

Start

Address (Hex)	Address	Data
1F72	8050	2
1F73	8051	5
1F74	8052	7
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Registers		Flag
A	01	S 0
BC	04 00	Z 0
DE	00 00	AC 0
HL	00 00	P 0
PSW	00 00	C 0
PC	42 0C	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 8000
2 MOV B, A
3 LDA 8001
4 SUB B
5 STA 8002
6 RST 1
```

Start 8000

Address (Hex)	Address	Data
1F40	8000	4
1F41	8001	5
1F42	8002	1
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Registers		Flag
A	02	S 0
BC	04 00	Z 1
DE	00 0A	AC 0
HL	00 14	P 1
PSW	00 00	C 0
PC	42 16	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 • LDA 2200
2 MOV E,A
3 MVI D,00
4 LDA 2201
5 MOV C,A
6 LXI H,0000
7 BACK: DAD D
8 DCR C
9 JNZ BACK
10 SHLD 2202
11 HLT
```

Start 2200 OK

Address (Hex)	Address	Data
0898	2200	10
0899	2201	2
089A	2202	20
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

Registers		Flag
A	03	S 1
9C	02 03	Z 0
DE	00 00	AC 0
HL	00 00	P 0
SW	00 00	C 1
PC	42 1B	
SP	FF FF	
-Reg	00	

Decimal - Hex Conversion

Decimal

Hex

0

To Hex

To Dec

O Ports

-

+

00

Update Port Value

Memory

-

+

00

Update Memory

Load me at

```
1  START:  NOP
2
3      LDA 8500
4      MOV B,A
5
6      LDA 8501
7      MVI C,00H
8
9  LOOP:  CMP B
10         JC END
11
12         SUB B
13         INR C
14         JMP LOOP
15
16  END:   STA 8502
17         MOV A,C
18         STA 8503
19
20        HLT
```

Data Stack Keypad M

Start 8500		
Address (Hex)	Address	Data
2134	8500	2
2135	8501	6
2136	8502	0
2137	8503	3
2138	8504	0
2139	8505	0
213A	8506	0
213B	8507	0
213C	8508	0
213D	8509	0
213E	8510	0
213F	8511	0
2140	8512	0
2141	8513	0

Line No	Assembler Message
0	Program assembled successfully



Registers			Flag	
A	09		S	0
BC	04 00		Z	0
DE	00 00		AC	0
HL	00 00		P	1
PSW	00 00		C	0
PC	42 17			
SP	FF FF			
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 3050
2 MOV B,A
3 LDA 3051
4 ADD B
5 STA 3052
6 LDA 3053
7 MOV B,A
8 LDA 3054
9 ADC B
10 STA 3055
11 HLT
```

Start 3050 OK

Address (Hex)	Address	Data
0BEA	3050	2
0BEB	3051	3
0BEC	3052	5
0BED	3053	4
0BEE	3054	5
0BEF	3055	9
0BF0	3056	0
0BF1	3057	0
0BF2	3058	0
0BF3	3059	0
0BF4	3060	0
0BF5	3061	0
0BF6	3062	0
0BF7	3063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Search



ENG IN



14:05 10-08-2026

00

S 0

00 00

Z 1

00 00

AC 0

42 22

P 1

00 0A

C 0

0

To Dec

- + 00

Update Port Value

- + 00

Update Memory

1 LHL D 2050

2 SPHL

3 LHL D 2052

4 XCHG

5 LXI H,0000H

6 LXI B,0000H

7 AGAIN: DAD SP

8 JNC START

9 INX B

10 START: DCX D

11 MOV A,E

12 ORA D

13 JNZ AGAIN

14 SHLD 2054

15 MOV L,C

16 MOV H,B

17 SHLD 2055

18 HLT

Start 2050

Address (Hex)	Address	Data
0802	2050	10
0803	2051	0
0804	2052	5
0805	2053	0
0806	2054	50
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No	Assembler Message
0	Program assembled successfully

00

00

00

00

00

1A

FF

00

Flag

S 1

Z 0

AC 0

P 0

C 1

Hex Conversion

Hex

0

To Dec

+ 00

ate Port Value

+ 00

ate Memory

Load me at

1 LDA 8501

2 MOV B,A

3

4 LDA 8500

5

6 MVI C,00

7

8 LOOP: CMP B

9 JC DONE

10 SUB B

11 INR C

12 JMP LOOP

13

14 DONE: STA 8503

15 MOV A,C

16 STA 8502

17 RST 1

Data

Stack

Keypad

Men

Start 8050

Address (Hex)	Address	Data
1F72	8050	2
1F73	8051	20
1F74	8052	10
1F75	8053	2
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No	Assembler Message
0	Program assembled successfully



Registers			Flag	
A	18		S	1
BC	18	00	Z	0
DE	00	00	AC	0
HL	00	00	P	0
PSW	00	00	C	1
PC	42	10		
SP	FF	FF		
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 2050
2 MOV B,A
3 LDA 2051
4 CMP B
5 JNC STORE
6 MOV A,B
7 STORE: STA 2052
8 HLT
```

Start

Address (Hex)	Address	Data
0802	2050	24
0803	2051	12
0804	2052	24
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

File
Reset
Assembler
Debug
Help

Registers

A	19
BC	19 03
DE	0C 22
HL	1F 41
PSW	00 00
PC	42 0F
SP	FF FF
Int-Reg	00

Flag

S	0
Z	0
AC	0
P	1
C	0

Load me at

```

1 LXI H, 8000
2 MOV B,M
3 INX H
4 MOV A,M
5 CMP B
6 JC STORE
7 MOV A,B
8
9 STORE: STA 8010
10 HLT
        
```

Decimal - Hex Conversion

To Hex
To Dec

I/O Ports

Update Port Value

Memory

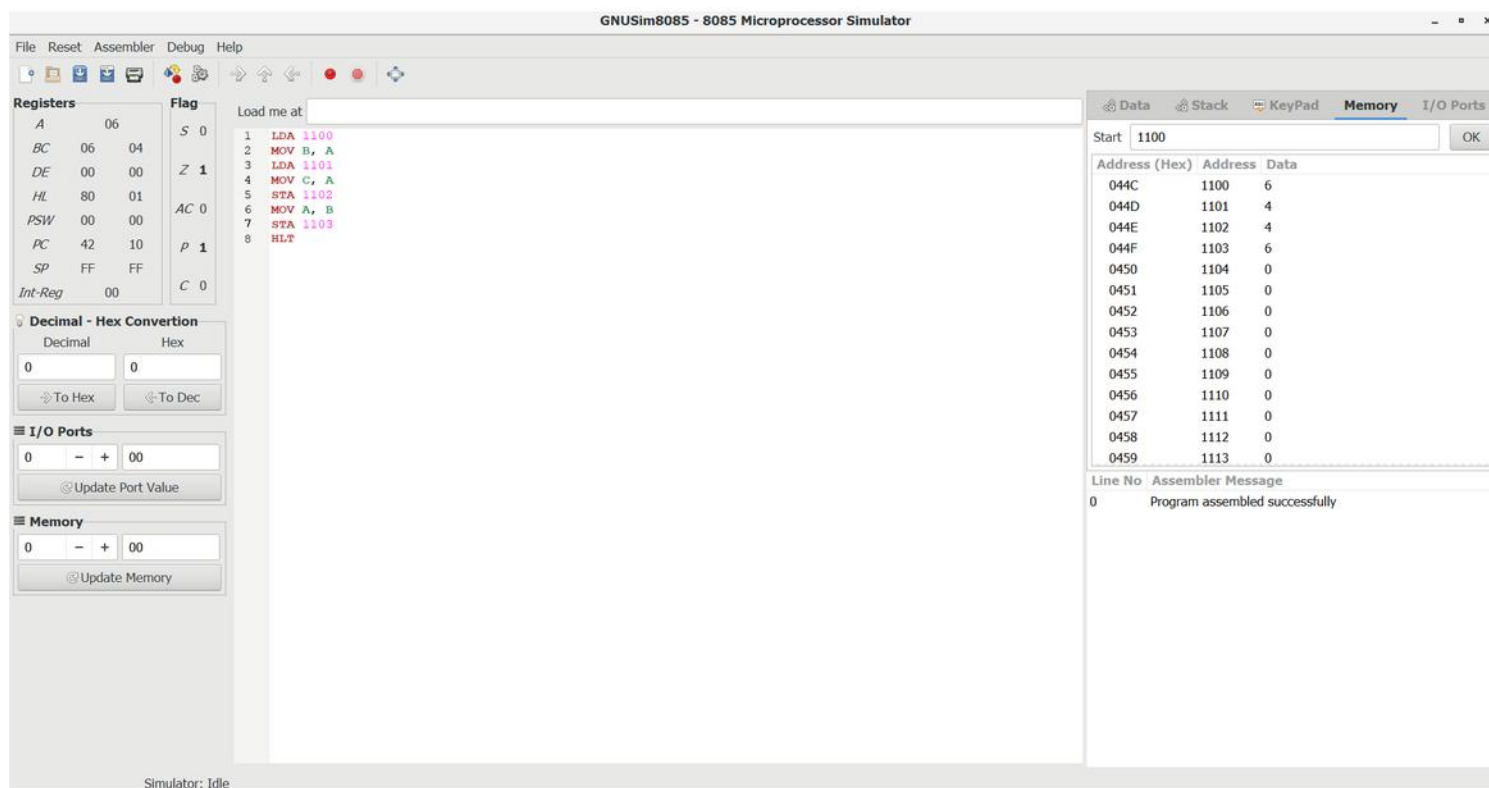
Update Memory

Data
Stack
Keypad
Memory
I/O Port

Start OK

Address (Hex)	Address	Data
1F40	8000	25
1F41	8001	35
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	25
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message
0 Program assembled successfully



Registers

A	DC	S	0
BC	00 00	Z	0
DE	00 00	AC	0
HL	00 00	P	0
PSW	00 00	C	0
PC	42 08		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

0

-

+

00

Update Memory

Load me at

1

LDA

8000

2

CMA

3

STA

8001

4

HLT

Data

Stack

KeyPad

Memory

I/O Ports

Start

8001

OK

Address (Hex)	Address	Data
1F41	8001	220
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0
1F4F	8015	0
1F50	8016	0

Line No	Assembler Message
0	Program assembled successfully

Registers

A	FB	S	1
BC	00 00	Z	0
DE	00 00	AC	0
HL	00 00	P	0
PSW	00 00	C	0
PC	42 0A		
SP	FF FF		
Int-Reg	00		

Flag

S	1
Z	0
AC	0
P	0
C	0

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

1 LDA 3000H
2 CMA
3 ADI 01H
4 STA 3001H
5 HLT

Data Stack KeyPad Memory I/O Ports

Start 3001h OK

Address (Hex)	Address	Data
3001	12289	251
3002	12290	0
3003	12291	0
3004	12292	0
3005	12293	0
3006	12294	0
3007	12295	0
3008	12296	0
3009	12297	0
300A	12298	0
300B	12299	0
300C	12300	0
300D	12301	0
300E	12302	0
300F	12303	0
3010	12304	0

Line No Assembler Message

0 Program assembled successfully



Registers		Flag
A	16	S 0
BC	00 00	Z 1
DE	00 00	AC 1
HL	00 00	P 1
PSW	00 00	C 0
PC	42 13	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 8050H
2 ANI 01
3 JZ LOOP1
4 MVI A,11
5 JMP LOOP2
6 LOOP1: MVI A,22
7 LOOP2: STA 8051
8 HLT
```

Start

Address (Hex)	Address	Data
1F72	8050	20
1F73	8051	22
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Registers		Flag
A	16	S 1
BC	00 00	Z 0
DE	00 00	AC 1
HL	00 00	P 0
PSW	00 00	C 0
PC	42 12	
SP	FF FF	
Int-Reg	00	

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 MVI A,9AH
2 ANI 80H
3 JZ NEG
4 MVI A,22
5 JMP ST0
6 NEG: MVI A,11
7 ST0: STA 8051
8 HLT
```

Start 8050			OK
Address (Hex)	Address	Data	
1F72	8050	15	
1F73	8051	22	
1F74	8052	0	
1F75	8053	0	
1F76	8054	0	
1F77	8055	0	
1F78	8056	0	
1F79	8057	0	
1F7A	8058	0	
1F7B	8059	0	
1F7C	8060	0	
1F7D	8061	0	
1F7E	8062	0	
1F7F	8063	0	

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Registers		Flag	
A	04	S	0
BC	00 00	Z	1
DE	00 00	AC	0
HL	1F 42	P	1
PSW	00 00	C	0
PC	42 1D		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LXI H,8000
2 MOV C,M
3 DCR C
4 LOOP3: MOV D,C
5 LXI H,8001
6 LOOP2: MOV A,M
7 INX H
8 CMP M
9 JC LOOP1
10 MOV B,M
11 MOV M,A
12 DCX H
13 MOV M,B
14 INX H
15 LOOP1: DCR D
16 JNZ LOOP2
17 DCR C
18 JNZ LOOP3
19 HLT
```

Start 8000 OK

Address (Hex)	Address	Data
1F40	8000	3
1F41	8001	0
1F42	8002	4
1F43	8003	18
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfully

File Reset Assembler Debug Help



Registers			Flag	
A	78		S	0
BC	00 06		Z	1
DE	00 78		AC	0
HL	1F 42		P	1
PSW	00 00		C	0
PC	42 1B			
SP	FF FF			
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LDA 2001
2 MOV B,A
3 MVI C,#01
4 MVI E,#01
5 LOOP: MOV D,C
6 MVI A,80H
7 LP: ADD E
8 DCR D
9 JNZ LP
10 MOV E,A
11 INR C
12 DCR B
13 JNZ LOOP
14 MOV A,E
15 STA 2010
16 HLT
```

Data Stack Keypad **Memory** I/O Ports

Start 2001			OK
Address (Hex)	Address	Data	
07D1	2001	5	
07D2	2002	6	
07D3	2003	3	
07D4	2004	2	
07D5	2005	5	
07D6	2006	0	
07D7	2007	0	
07D8	2008	0	
07D9	2009	0	
07DA	2010	120	
07DB	2011	0	
07DC	2012	0	
07DD	2013	0	
07DE	2014	0	

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

FileResetAssemblerDebugHelp

Registers

A00

BC0000

DE0003

HL8500

PSW0000

PC4217

SPFFFF

Int-Reg00

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LXI H, 8050H

2MOV C, M

3INX H

4MOV B, M

5DCR C

6

7LOOP: INX H

8MOV A, M

9CMP B

10JNC SKIP

11MOV B, A

12

13SKIP: DCR C

14JNZ LOOP

15

16LXI H, 8500H

17MOV M, B

18HLT

DataStackKeyPadMemoryI/O Ports

Start8051OK

Address (Hex)	Address	Data
1F73	8051	5
1F74	8052	2
1F75	8053	6
1F76	8054	8
1F77	8055	9
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0
1F80	8064	0

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A07

BC0700

DE0000

HL8054

PSW0000

PC4217

SPFFFF

Int-Reg00

Flag

S0

Z1

AC0

P1

C1

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LXI H, 8050H

2MOV A, M

3MOV B, A

4MVI C, 04H

5

6LOOP: INX H

7MOV A, M

8CMP B

9JC SKIP

10MOV B, A

11

12SKIP: DCR C

13JNZ LOOP

14

15MOV A, B

16STA 8055H

17HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

8050328485

8051328492

8052328504

8053328517

8054328525

8055328537

8056328540

8057328550

8058328560

8059328570

805A328580

805B328590

805C328600

805D328610

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help



Registers			Flag	
A	00		S	0
BC	4B 2D		Z	1
DE	B5 00		AC	0
HL	E1 00		P	1
PSW	00 00		C	0
PC	42 2D			
SP	FF FF			
Int-Reg	00			

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

0	-	+	00
Update Memory			

Load me at

```
1 LXI H,0000
2
3 MOV C,M
4
5 MVI B,00
6 INX H
7
8 MOV B,M
9 CMA
10 MOV E,A
11 MVI D,00FH
12 MOV A,B
13 CMA
14 MOV D,A
15 INX D
16 LXI H,0000
17 NEXT: DAD B
18 SHLD 0010
19 LOOP: DAD D
20 JNC SKIP
21 MOV A,H
22 ORA L
23 JZ EXIT
24 JMP LOOP
25 SKIP: LHLD 0010
26 JMP NEXT
27 EXIT: LHLD 0010
28 HLT
29
```

Data Stack Keypad Memory I/O Ports

Start 8000 OK

Address (Hex)	Address	Data
1F40	8000	45
1F41	8001	75
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	225
1F4C	8012	0
1F4D	8013	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

33°C
Mostly cloudy



ENG IN 10:50 31-08-2026

FileResetAssemblerDebugHelp

Registers

A13

BC132D

DEB500

HL1F41

PSW0000

PC421B

SPFFFF

Int-Reg00

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

DecimalHex

00

0

To Hex

To Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at

1LXI H,8000

2MOV A,M

3INX H

4MOV B,M

5LOOP: CMP B

6JZ STORE

7JC EXG

8SUB B

9JMP LOOP

10EXG: MOV C,B

11MOV B,A

12MOV A,C

13JMP LOOP

14STORE: STA 8009

15HLT

DataStackKeyPadMemoryI/O Ports

Start8000OK

Address (Hex)AddressData

1F40800038

1F41800119

1F4280020

1F4380030

1F4480040

1F4580050

1F4680060

1F4780070

1F4880080

1F49800919

1F4A80100

1F4B80110

1F4C80120

1F4D80130

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A78

BC0000

DE7800

HIL1F73

PSW0000

PC4210

SPFFFF

Int-Reg00

Flag

S0

Z1

AC0

P1

C0

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

0-+00

Update Memory

Load me at:

1LXI H,0050

2MOV B,M

3MVI D,01H

4FACT: CALL MUL

5DCR B

6JNZ FACT

7INX B

8MOV M,D

9HLT

10MUL: MOV E,B

11XRA A

12ML: ADD D

13DCR E

14JNZ ML

15MOV D,A

16RET

17HLT

DataStackKeyPadMemoryI/O Ports

Start8050OK

Address (Hex)AddressData

1F7280505

1F738051120

1F7480520

1F7580530

1F7680540

1F7780550

1F7880560

1F7980570

1F7A80580

1F7B80590

1F7C80600

1F7D80610

1F7E80620

1F7F80630

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle



Registers		Flag	
A	54	S	0
BC	44 2D	Z	0
DE	B5 00	AC	0
HL	08 04	P	0
PSW	00 00	C	0
PC	42 0E		
SP	FF FF		
Int-Reg	00		

Decimal - Hex Conversion

Decimal	Hex
0	0

I/O Ports

0	-	+	00
---	---	---	----

Memory

0	-	+	00
---	---	---	----

Load me at

```
1 LXI H,2050
2 MOV A,M
3
4 ADD A
5
6 MOV B, A
7 ADD A
8 ADD A
9 ADD B
10 INX H
11 ADD H
12 INX H
13 MOV M,A
14 HLT
15
```

Start 2050

Address (Hex)	Address	Data
0802	2050	34
0803	2051	0
0804	2052	84
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle



Search

